SAMYAK SIDDHARTH SANGHVI

samyakss83.github.io



ACADEMIC DETAILS

B.Tech in Computer Science & Engineering, Indian Institute of Technology Delhi **GPA:** 9.35 | **Dept. Rank:** 17

SCHOLASTIC ACHIEVEMENTS

- **Joint Entrance Exam Advanced (JEE) 2023:** Secured **All India Rank 83** amongst more than 1.5 Lakh candidates.
- **KVPY Scholar 2022:** Awarded the KVPY scholarship by securing **All India Rank 27** amongst more than 50000 students.
- **United Nations Round Table:** Invited to attend a Panel of 30+ students for a dialogue with **Deputy Secretary General**
- **INPhO 2023:** Among **top 35 students** from the country, selected to attend Orientation and Selection Camp, TIFR Mumbai.
- **APhO 2023:** Selected for national team of Asian Physics Olympiad 2023 to represent India at Ulaanbaatar, Mongolia.
- **Eightfold AI INNOV8 Challenge:** Selected as one of the **20 national finalists** in hackathon organized by Eightfold AI.
- **InterIIT Tech Meet 14.0 Patna:** Lead the team in the **Geospatial Intelligence** problem statement and won the **Gold Medal**.
  KLA AI Hack: Won **First Prize** for delivering the **best** image denoising solution under latency and resource constraints.

PUBLICATIONS & PATENTS

- **Attend What Matters: Leveraging Vision Foundational Models for Breast Cancer Classification Using Mammograms**
  IEEE International Symposium on Biomedical Imaging, 2026 | *(Prof. Chetan Arora)*
  - Generated ROIs from **Grounding DINO**, applied **cross-attention** between crops and full-breast, classified via an **MLP.**
  - Fine-tuned **DINOv2** and trained a classifier with **contrastive loss** for better embedding **alignment** and discrimination.
- **Towards Support-Free Printing in Extrusion-based Additive Manufacturing: Path Planning and Process Control**
  International Mechanical Engineering Congress and Exposition, 2025 | *(Prof. Sagar Sarkar)*
  - Created a pipeline to achieve graphics based multi-axis adaptive slicing to allow **support-less overhang printing**
  - Designed and implemented a method to **Identify, label, and segment mesh extrusions** using SDF to cut the Mesh.
  - Designed a **fault detection interrupt** pipeline via **deterministic computer vision** analysis to flag bumps and strings
- **SiDGen: Structure-informed Diffusion for Ligand Generation** | *(Prof. Tarak Karmakar)*
  - Designed **protein-conditioned ligand diffusion** with **Triangle Attention** for structure and **Cross Attention** for denoising.
  - Used **DiffPool soft coarsening** to capture **3D structural context** with **lower memory** and **better feature pooling**.

INTERNSHIPS

- **Temple Pvt. Ltd.** (July, 2026 - Present) : *Electrophysiological Signal Modeling*
  - Developing models and evals on **electrophysiological signals** for decoding **neurological** and **physiological metrics**.
  - Improved **cross-subject generalization** using **classical**, **deep learning** and **foundation models** optimized as per task.
  - Built **distributed GPU infrastructure** with **persistent storage**, **per-user SSH isolation** and **reproducible environments**.
- **Quadeye Securities, Quantitative Strategist** (May, 2026 - July, 2026) : *Pair Trading Strategies*
  - Worked on a **foundational model** for **pair trading strategies** spanning **high-** and **medium-frequency** trading regimes.
  - Used multiple flavours of attention as well as pretraining losses and novel feature interaction pipelines for robust predictions
- **Stylout AI, Founding Engineer** (May, 2025 - September, 2025) : *Virtual Tryon Pipeline and Recommendation System*
  - Developed **VITON pipeline** using DensePose, body poses, and **fine-tuned diffusion model** for image generation.
  - Built **recommendation engine** incorporating Exploration-Exploitation via **GCN embeddings** for fashion relationships.
- **FAIRPHILE INNOVATION LABS** (December 2024, - January, 2025) : *Efficient 3D reconstruction and segmentation*
  - Developed **fine-tuned masking models** with **SAM-2 and YOLO-v8** for object segmentation and **mask generation**.
  - Implemented custom **NeRF** and its pipeline to **reconstruct segmented point-cloud** from **sparse multi-view imagery**.

PROJECTS

- **Grounded Radiology Report Generation** | *(Prof. Chetan Arora)*
  - Proposed **Simpler**, a unified RRG model integrating **MedCLIP retrieval**, diagnostic conditioning, and **in-context reports**.
  - Built **phrase grounding annotations** using **MedGemma-27B**, linking reports to **image regions** and diagnostic conditions.
  - Introduced an **attention budget** with **phrase-attention supervision** to ground generated findings to visual regions.
- **Multi-View Consistent Report Generation for Mammography** | *(Prof. Chetan Arora)*
  - Proposed **MVCNet**, fusing CC, MLO views via **cross-attention**, beating SOTA VLMs by **4.4%** using **13$\times$ smaller model.**
  - Trained with **InfoNCE contrastive pretraining** for alignment, followed by **RAG fine-tuning** using top-K reports as context.
  - Leveraged **LLM-extracted findings** supervising a **multilabel classification** branch, reducing template **hallucinations**.
- **Natural Language Interpretation of Satellite Imagery** | **(InterIIT Tech Meet 14.0)** (Novermber 2025–December 2025)
  - Used a composition of **Qwen3 VL** and **SAM3** with a Graph-based **Semantic mask merging** algorithm to localize objects
  - Created a **Mixture-of-Adapters** architecture to support **SAR**, **IR** images, added **ReAct Tool calling** for Geometric analysis.
- **General Vision Language Action Navigator** | **(Robotics Club)** (September 2025–December 2025)
  - Integrated **Grounding DINO** and **Depth Anything V2** for **obstacle detection** while maintaing localization using odometry.
  - Leveraged **Qwen2.5 VL** and **SAM2** for **destination and task reasoning**, and **safe actions** when obstacles were detected.
- **Nil Proxy Pro - Portable Biometric Attendance System** (May 2024 – Aug 2024)
  - Implemented a **OS** and **micro-kernel** with proprietary drivers for fingerprint sensor and RTC, with WiFi and ESP32 support.
  - Integrated **secure cloud infrastructure** to store, upload, and manage data with high availability, while being power efficient.